

Universal 1-D Analytical-Signal Encoder

100k-molecule, 12-epoch scaling gate and experimental-transfer evaluation

OUTCOME: STOP EPOCH SCALING - RETAIN EPOCH 8 AND REDESIGN CROSS-TECHNIQUE LEARNING

Scaling from the repaired 10k checkpoint to 100k molecules produced a small simulated-probe gain, but extra training after epoch 8 did not help. Exact 800-label macro-AUPRC is 0.1889 at epoch 8 and 0.1889 at epoch 12: paired delta +0.000011, bootstrap 95% CI -0.000397 to +0.000421, versus the predeclared +0.003 continuation bar.

The central negative result is experimental transfer. On MassBank v2, epoch 8 is only +0.0010 above random for positive mode and +0.0007 for negative mode; both confidence intervals cross zero. It is significantly worse than the earlier repaired checkpoint by -0.0037 and -0.0047, respectively.

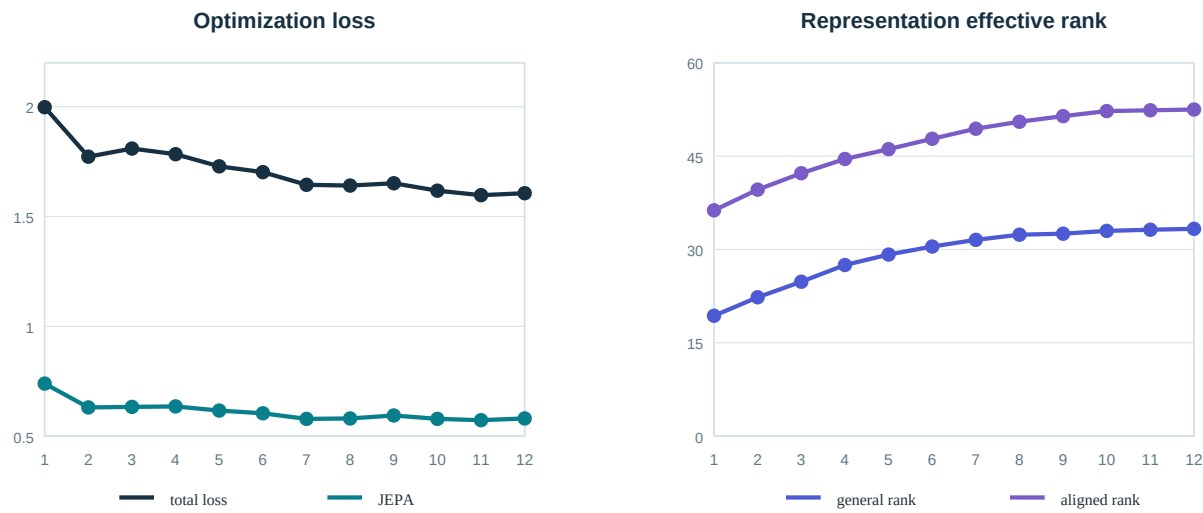
0.1889 epoch-8 1% AUPRC	+0.000011 epoch 12 vs 8	+0.0057 vs repaired 10k	NO-GO continue epochs
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Scope and provenance

Corpus	100,000 molecules; 900,000 simulated IR, 1H/13C NMR, and MS/MS spectra
Split	80/10/10 Bemis-Murcko scaffold split; molecule acquisitions remain together
Model	Shared 12-layer Transformer; 384-D general and 256-D aligned embeddings
Hardware	Apple M4 Max Studio, 36 GB unified memory; PyTorch MPS only
Training	12 epochs; 39,996 steps; 9.86 hours; status complete
Primary audit	10k/3k/3k cached molecules; 800 labels/family; five fixed seeds
Experimental	MassBank 2026.03 v2; 91,611 MS2 records; 10,902 molecules
Selected checkpoint	2987ba8a69745f2dd29a37cad940f710fdf304039a568b883675e22c83dbb660

1. Training dynamics

Optimization remained numerically stable through all 12 epochs. The model never crossed the collapse threshold, and both general and aligned effective rank increased rather than contracting. Falling training loss is therefore real optimization progress, but it is not evidence that downstream chemistry continues improving.



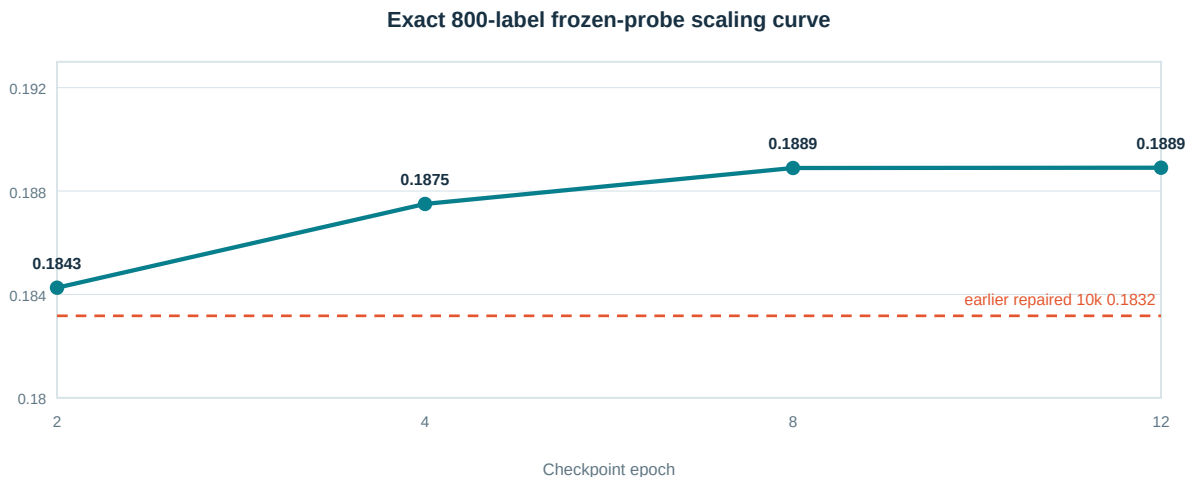
Diagnostic	Epoch 1	Epoch 8	Epoch 12	Reading
Total loss	1.998	1.641	1.606	Continues falling slowly
JEPA loss	0.739	0.581	0.581	Latent prediction improves
General rank	19.3	32.4	33.3	No collapse; compact space
Aligned rank	36.3	50.5	52.5	Broad aligned projection

What good training diagnostics would look like

Healthy ranks and decreasing JEPA loss are necessary, not sufficient. The decisive numbers are label-efficient probes, cross-family retrieval, and sim-to-real transfer. A genuinely improving universal encoder would show a downstream curve that rises with epoch, at least +0.003 from epoch 8 to 12, without family tradeoffs; that did not occur.

2. Exact 1% functional-group scaling curve

The exact protocol uses 800 labeled molecules per acquisition family, corresponding to 1% of the original 80,000-molecule training partition. Every checkpoint uses identical 10k train, 3k validation, and 3k test molecule subsets, with five fixed probe seeds and validation-tuned linear probes.

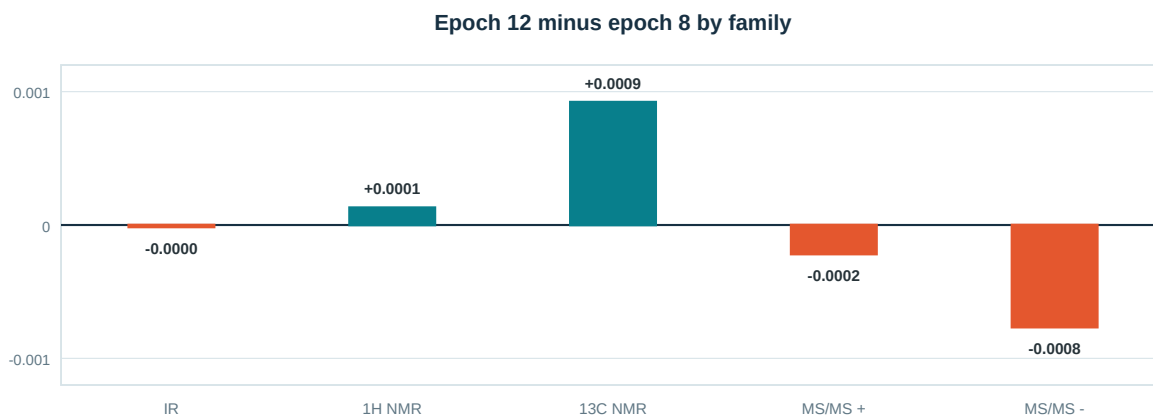


Checkpoint	Mean macro-AUPRC	Delta vs prior	Delta vs 10k baseline
Epoch 2	0.184259	-	+0.001087
Epoch 4	0.187501	+0.003241	+0.004329
Epoch 8	0.188892	+0.001391	+0.005720
Epoch 12	0.188902	+0.000011	+0.005730

Interpretation

Most useful improvement arrives by epoch 4. Epoch 8 reaches 0.1889, a +0.0057 gain over the earlier repaired 10k checkpoint. Epoch 12 is numerically higher by only +0.000011; the confidence interval is tightly centered on zero. More epochs under the current objective are not the limiting resource.

3. Family-wise continuation gate



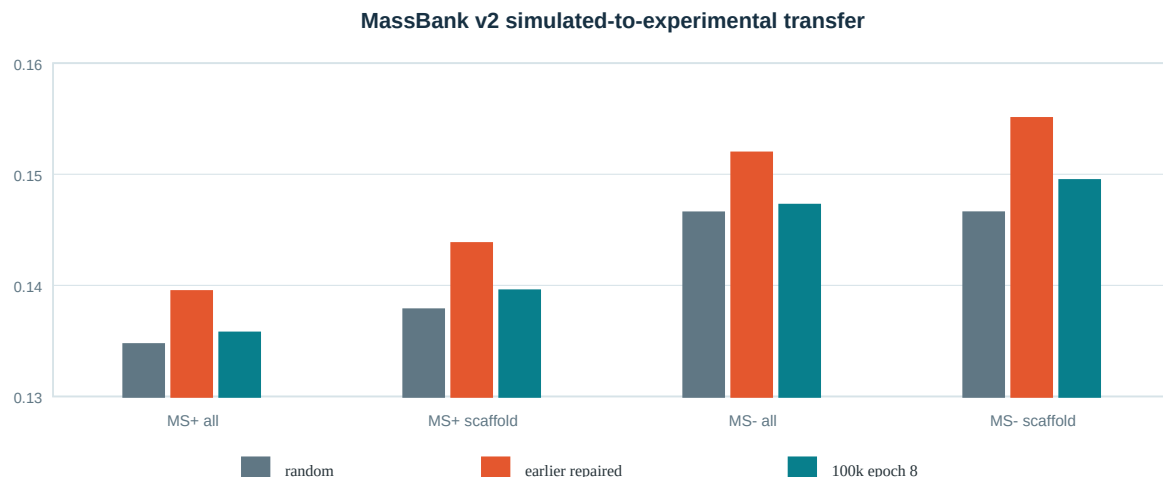
Family	Epoch 8	Epoch 12	Delta	Gate reading
IR	0.249868	0.249853	-0.000015	flat
1H NMR	0.190339	0.190469	+0.000130	flat
13C NMR	0.194227	0.195149	+0.000922	flat
MS/MS +	0.154777	0.154559	-0.000218	flat
MS/MS -	0.155246	0.154480	-0.000766	flat

No family violates the -0.02 regression floor; the largest decline is MS/MS negative at -0.00077. That is reassuring for stability, but it does not rescue the continuation decision. IR is flat, 1H NMR is flat, 13C NMR gains only +0.00092, and both MS modes decline slightly.

Continuation criterion	Threshold	Observed	Decision
Epoch 12 minus epoch 8 mean	$\geq +0.003$	+0.000011	FAIL
Bootstrap 95% CI	Support positive gain	[-0.000397, +0.000421]	FAIL
Worst family regression	> -0.020	-0.000766	PASS
Collapse	None	None	PASS

4. Experimental MassBank transfer

The selected epoch-8 encoder was applied to the licensed MassBank 2026.03 v2 store. Linear probes were trained on simulated MS embeddings and tested on experimental MS spectra. Exact-molecule-unseen and scaffold-unseen strata guard against direct overlap. The matched random-tokenizer control and earlier repaired 10k checkpoint use the same records and five seeds.



Mode / stratum	Random	Earlier repaired	100k epoch 8	vs repaired	vs random
MS+ all	0.1347	0.1395	0.1357	-0.0037	+0.0010
MS+ exact unseen	0.1350	0.1396	0.1359	-0.0037	+0.0009
MS+ scaffold unseen	0.1378	0.1438	0.1395	-0.0043	+0.0017
MS- all	0.1465	0.1519	0.1472	-0.0047	+0.0007
MS- exact unseen	0.1467	0.1519	0.1472	-0.0047	+0.0005
MS- scaffold unseen	0.1465	0.1550	0.1494	-0.0056	+0.0029

Interpretation

Relative to random, the 100k model gains only +0.0010 on MS+ and +0.0007 on MS- overall; both intervals include zero. Relative to the earlier repaired checkpoint it loses -0.0037 and -0.0047, with intervals excluding zero. Simulated label efficiency and experimental transfer have therefore diverged.

5. Nonlinear embedding maps

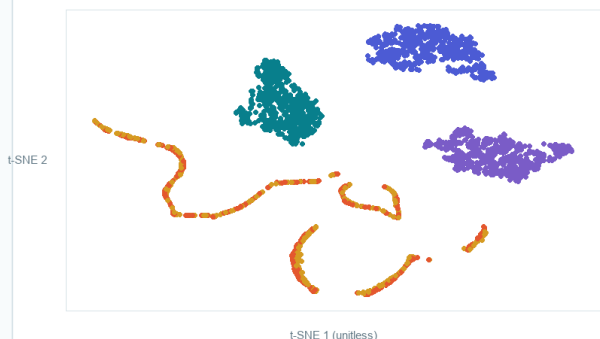
A balanced scaffold-held-out sample contains 600 molecules and one spectrum per family. General and aligned embeddings were standardized, reduced to 50 principal components, then independently fit with t-SNE. These maps are descriptive: local neighborhoods are meaningful; global island size and distance are not.

100k scaling pilot: held-out embedding neighborhoods

600 scaffold-held-out molecules · 3,000 balanced spectra · independent t-SNE fits after 50-D PCA

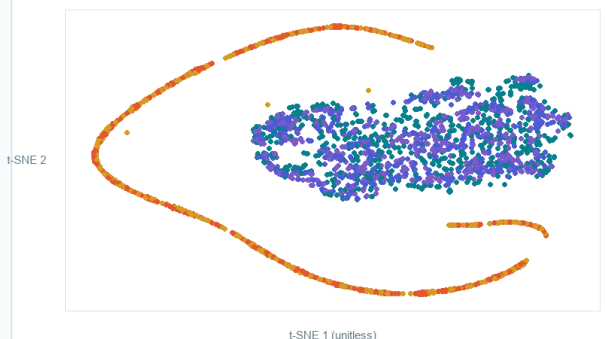
General embedding · acquisition

kNN acquisition agreement 79.9%



Aligned projection · acquisition

kNN acquisition agreement 42.6%



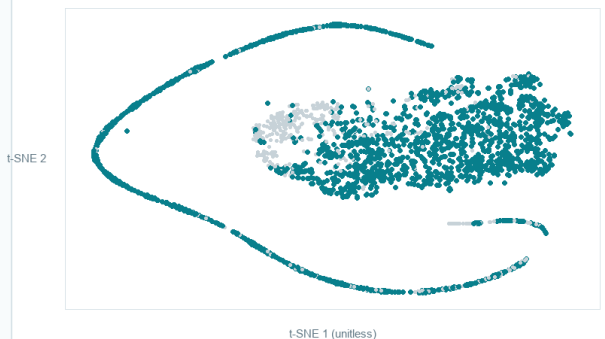
General embedding · Arene

kNN label Jaccard 0.406



Aligned projection · Arene

kNN label Jaccard 0.422



● IR

● ¹H NMR

● ¹³C NMR

● MS/MS +

● MS/MS -

Arene present

absent

t-SNE preserves local neighborhoods, not global distances or island sizes; quantitative claims use probes and retrieval.

The general embedding has 79.9% acquisition agreement among 15-nearest neighbors. The aligned head reduces this to 42.6%, while label Jaccard rises only from 0.406 to 0.422. Alignment mixes modalities, but does not yet create a strongly chemistry-organized universal space.

6. From signals to embeddings

Three held-out molecules are followed through all five acquisition families. Each trace is the canonical 4,096-bin signal; the orange line is patch-token energy, and the red/blue strip is the normalized 256-D aligned vector. This makes the representation path inspectable without implying that individual latent dimensions have direct chemical meanings.

Held-out spectra through the universal encoder

4,096 intensity bins $\rightarrow$ 128 patch tokens $\rightarrow$ 384-D general + 256-D aligned embeddings



Signals are robust-scaled and axes are stored in canonical increasing order. Patch energy is the token-vector norm; the barcode shows normalized aligned components.

7. Mechanistic diagnosis

The project is not limited by embedding rank or optimizer convergence. The failure is geometric and domain-specific. IR and NMR share dense spectral structure, while MS is sparse and precursor-conditioned. A single global alignment objective can reduce modality separation without learning the same chemistry-relevant invariances across both regimes.

Observation	Evidence	Implication
Healthy rank	General rank 33.3; aligned rank 52.5	Not representation collapse
Epoch plateau	Epoch 8 to 12 +0.000011	More time is low value
Modality structure	General kNN acquisition agreement 79.9%	Strong shortcut remains
Alignment helps mixing	Aligned agreement 42.6%	But chemistry organization remains weak
Sim-to-real divergence	MassBank vs repaired: MS+ -0.0037, MS- -0.0047	Simulation-specific features strengthened

What this rules out

The result rules out the simplest hypothesis that larger data and longer training alone will unlock a universal representation. It does not rule out a shared backbone. Instead, it argues for structured technique grouping, explicit sim-to-real invariance, and stronger chemistry-bearing objectives.

8. Decision and next experiment

SCALING DECISION: STOP AT EPOCH 8

RESEARCH DECISION: CONTINUE WITH GROUPED BRIDGES AND EXPERIMENTAL DOMAIN ALIGNMENT

Priority	Experiment	Success signal
1	Separate spectroscopy and MS alignment bridges on the shared backbone	Improve IR/NMR without losing MS
2	Add small shared chemistry bottleneck with gradient-conflict monitoring	Positive cross-bridge retrieval and probe gain
3	Mix a bounded quantity of MassBank experimental MS during pretraining	Experimental gain over repaired 10k across seeds
4	Use acquisition-specific augmentations and intensity transforms	Robust sim-to-real transfer without shortcut growth
10k gate	Run the architectural test at 10k before scaling	+0.010 spectroscopy mean; MS $\geq$ 0; family floor -0.005

Why not add more techniques yet

Adding HPLC, PXRD, chromatography, or more analytical families now would increase heterogeneity before the model can bridge the existing spectroscopy/MS divide. The most informative next step is architectural and domain-alignment work on the current five families, not broader corpus expansion.

What a successful next pilot should achieve

Metric	Minimum	Compelling
800-label mean macro-AUPRC	+0.010 over epoch-8 control	+0.030 over best specialist/MOMENT
Family coverage	4 of 5 improve; none below -0.005	All 5 improve
MassBank transfer	Positive delta vs repaired, CI excluding zero	Gain in both ion modes and scaffold-unseen
Cross-family retrieval	R@1 > 0.10	R@1 > 0.20 with positive margins
Shortcut diagnostic	Aligned modality accuracy falls	Chemistry neighborhoods improve simultaneously

Appendix. Reproducibility and limitations

Item	Value
Run	pilot-fixed-maskrepair-100k-12ep
W&B;	https://wandb.ai/anelli/jepanalytics/runs/pilot-fixed-maskrepair-100k-12ep-v1
Epoch-8 checkpoint SHA-256	2987ba8a69745f2dd29a37cad940f710fdf304039a568b883675e22c83dbb660
Epoch-12 checkpoint SHA-256	b15ffcebe75afe375a0087420fdd04c098a7f21dd761f1800b9de5cd3855ffde
Dataset manifest SHA-256	3ee237af45f4333afd00a5d80b67b97b78482dbf53989e03e20daef4bdc73d02
Split SHA-256	52c6a78c933ae868eb6f0561075f71f0fc2c46e8ac4b8ae69dce30df9881a1e5
Evaluation subsets	10,000 train; 3,000 validation; 3,000 test molecules; seed 20260810
Probe seeds	11, 17, 23, 31, 47
MassBank	Release 2026.03 v2; 91,611 records; 10,902 molecules
Report generated	2026-08-15

Limits on inference

There is one pretraining seed. Five probe seeds measure labeled-subsample and probe variation, not pretraining variation. The +0.0057 simulated gain over the earlier repaired checkpoint is not compared here against complete MOMENT and single-technique specialist baselines, so it does not satisfy the full project go criterion. SpecTeach remains blocked by the proprietary Mnova container format; MassBank is the completed experimental signal-level validation.

Visual-map limits

The two t-SNE maps are fit independently and cannot be compared axis by axis. Trustworthiness is 0.997 for the general map and 0.990 for the aligned map, but global island distances remain non-metric. Functional-group labels are multilabel; the category panel shows presence versus absence for one selected SMARTS class.

Data and licensing

The approved NeurIPS multimodal spectroscopy corpus is used under the recorded CDLA-Sharing-1.0 audit entry. MassBank records were filtered to compatible record-level licenses. This report redistributes neither source nor derived spectra.